WellSense AIoT & System Product Hackathon

Build a wellness startup product prototype with human sensing, edge AI, dashboard, and feedback.

UNO Q main hub

Nano wearable node

Wellness cue

Briefing deck for all teams: what to build, how to find ideas, and how judging works.

Super AI Engineer Season 6 On-Site Week 3

CHALLENGE

Build a wellness product prototype, like a startup would.

Do not present only a sensor demo. Present a product that solves a real user pain point.

pain point → multi-sensor input → insight → dashboard → feedback → product value

Product question

Who pays or adopts?
What repeated pain exists?
Why is this better than current behavior?

Prototype proof

Live signal or state
Dashboard output
Feedback triggered by logic

Safety boundary

Wellness awareness only
No diagnosis
No medical decision claims

UNO Q is the system brain; Nano is the sensing companion.

Teams can add sensors, Modulino, actuators, and other supporting devices.

Nano 33 BLE Sense
wearable sensing node

→

UNO Q
main edge hub
logging + inference + dashboard

→

User
feedback

Nano role

Motion, gesture, sound, environment, low-power collection, optional TinyML.

UNO Q role

Bridge data, process signals, infer state, serve dashboard, control feedback.

Must explain

What Nano measures and how its data affects insight or feedback.

SYSTEM FLOW

A strong demo makes the full loop visible.

Judges should see data enter, become a state, appear on a dashboard, and trigger feedback.

1
Input

Sensors + context

2
Processing

Features + filtering

3
State

normal / warning / action

4
Dashboard

live signal + state

5
Feedback

LED / vibration / prompt

Best demos are not just dashboards. They prove that feedback reacts to live logic.

MVP

Minimum viable prototype: make this loop work first.

This is the baseline teams should finish before adding extra features.

- ✓ Nano sends at least 1 sensor signal to UNO Q or the system
- ✓ UNO Q combines it with at least 1 more signal or context
- ✓ Logic/model maps data into at least 3 states
- ✓ Dashboard shows at least 2 live or near-live outputs
- ✓ Feedback is triggered by logic, not only manual control
- ✓ Demo clearly shows input → state → dashboard → feedback
- ✓ Team explains how the insight helps the user pain point
- ✓ System is for wellness awareness, not medical decision-making

IDEA TRACKS

Choose a real user problem, then select signals that prove it.

Teams may invent their own concept; these tracks show feasible directions.

Posture & Focus Coach

Posture, tilt, distance, light, inactivity → better timing for work/focus reminders.

Calm Work Companion

Movement, sound, light, self-report → calm mode and self-regulation cues.

Movement Safety

IMU, sudden motion, orientation, proximity → risk cue with fewer false alarms.

Micro-Break Assistant

Desk + wearable context → break timing, hydration, posture, work rhythm.

Wellness Insight Dashboard

Multiple signals + event log → personal pattern and behavior awareness.

Good track test: Can the team explain why each signal matters to the user?

DELIVERABLES

Submit the product, the story, and enough evidence to judge it.

Think like a startup: product demo, user proof, system proof, and business path.

Prototype demo

Working sensor, processing, dashboard, and feedback flow.

Architecture one-pager

UNO Q, Nano, sensors, data flow, logic, dashboard, feedback.

Dashboard + logic

Live outputs plus a short explanation of model/rule choices.

Business canvas

Target user, pain point, current alternative, value proposition, channel, cost/revenue.

Pitch + video

5-7 slide pitch deck and 3-minute demo video.

Code + BOM

Repository/source package and complete sensor/device list.

PITCH STORY

A 5-7 slide pitch should move from problem to proof.

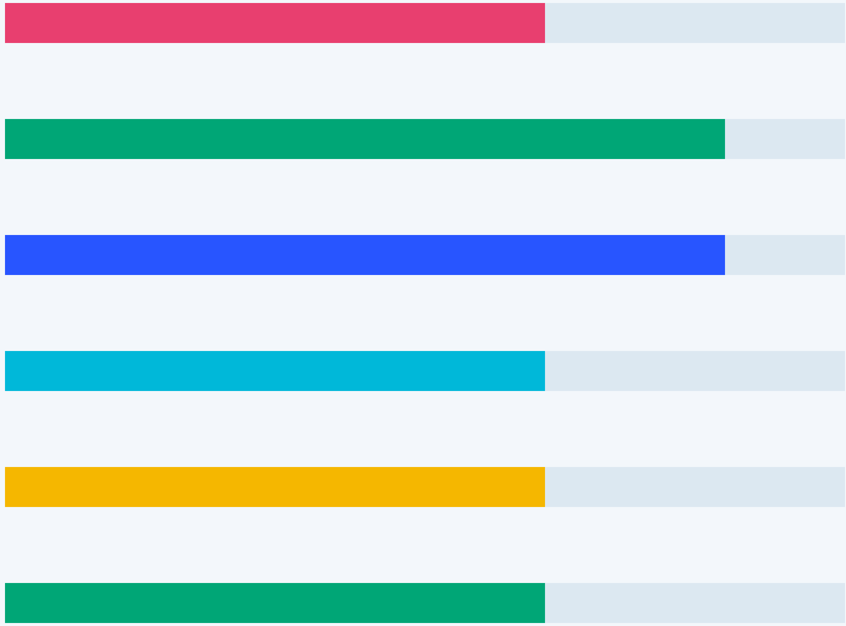
Keep the format free, but make sure judges can follow the logic quickly.

1. Problem & Pain Point
2. Proposed Solution
3. Multi-Sensor System
4. AI / Logic Insight
5. Dashboard & HCI
6. Business Canvas
7. Demo & Next Step

Judging is easier when every slide answers: What changed for the user because your system exists?

Rubric is now six clear buckets worth 100 points.

The largest weights reward system design, multi-sensor insight, and useful logic.



Problem & User Value	**15**
System Architecture & Hardware Use	**20**
Multi-Sensor Insight & AI Logic	**20**
Dashboard & Feedback Interaction	**15**
Prototype Quality, Maintainability & Low-Power	**15**
Business Canvas, Demo & Storytelling	**15**

100 points

Keep the product in wellness, not medicine.

The system may suggest cues, reminders, and awareness. It must not diagnose, treat, or replace professional judgment.

Safe words

wellness cue
behavior awareness
self-monitoring
support / reminder / guide

Risky words

diagnosis
treatment
medical decision
stress detection as fact

Safe demo

input → state → dashboard → feedback
with a clear disclaimer

Say: awareness and guidance. Do not say: diagnosis or treatment.

Treat the prototype like a startup product.

A strong team explains not only how it works, but why someone would use it repeatedly.

Target user

Pick a specific user group: students, desk workers, caregivers, athletes, makers, office teams.

Pain point

Name a repeated problem, not a vague wellness theme. Example: loses focus after long sitting.

Current alternative

What do users do now? Ignore it, use phone reminders, manual logging, ask someone to check.

Value proposition

What new insight or behavior change does the product create with sensors and feedback?

MVP proof

Show one live product loop first. Extra features matter only after the loop works.

Business path

Who adopts it, where it is used, what it costs, and how it could become a service.

PRODUCT EXAMPLES

Examples should sound like products, not sensor lists.

Each concept needs user, signal, insight, feedback, and a plausible business angle.

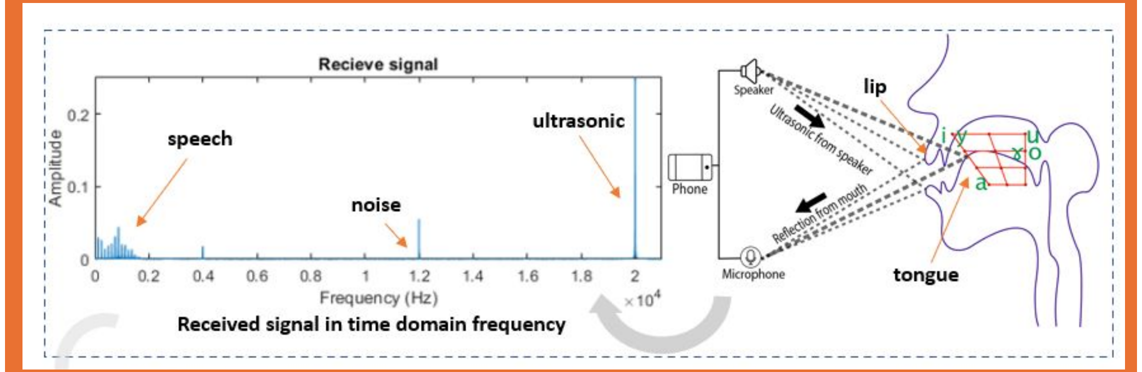
Product	User	Signals	Insight	Feedback	Business
Focus Coach for students	student during study session	posture + inactivity + light	focus drops after long still posture	gentle break cue	school / tutoring bundle
Desk Micro-Break Assistant	office worker	distance + posture + work rhythm	unhealthy sitting pattern	LED / prompt reminder	B2B wellness kit
Movement Awareness Companion	older adult / caregiver	IMU + sudden motion + context	movement risk cue, not fall diagnosis	alert to user/caregiver	home wellness service
Calm Work Companion	stressed knowledge worker	sound + motion + self-report	environment may trigger overload	breathing or light cue	subscription / workplace benefit

Medical-risk example: say “movement risk cue” or “awareness companion,” not “fall detector for diagnosis.”

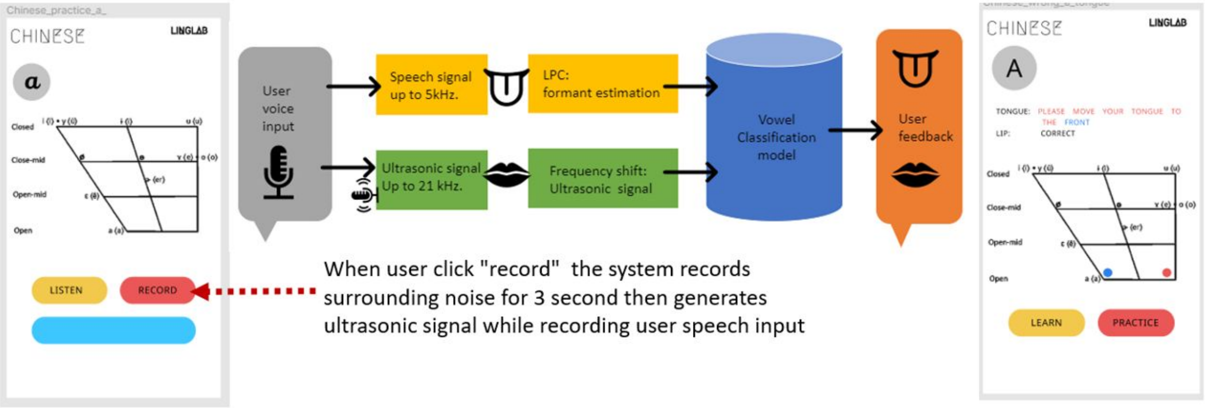
Articulation Sensing & Transparency

A Multifaceted Approach Advancing HCI Applications via Smartphones

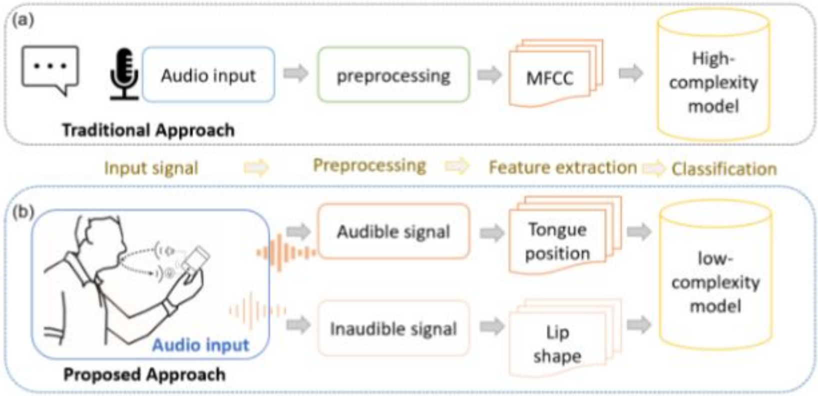
to sense articulation by speech and
the **Doppler effect of the ultrasonic signal**



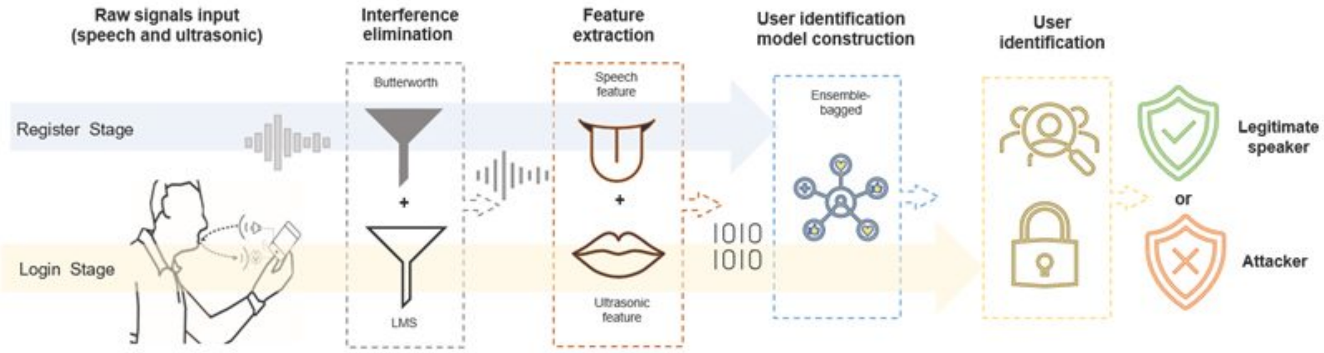
2. Pronunciation Training



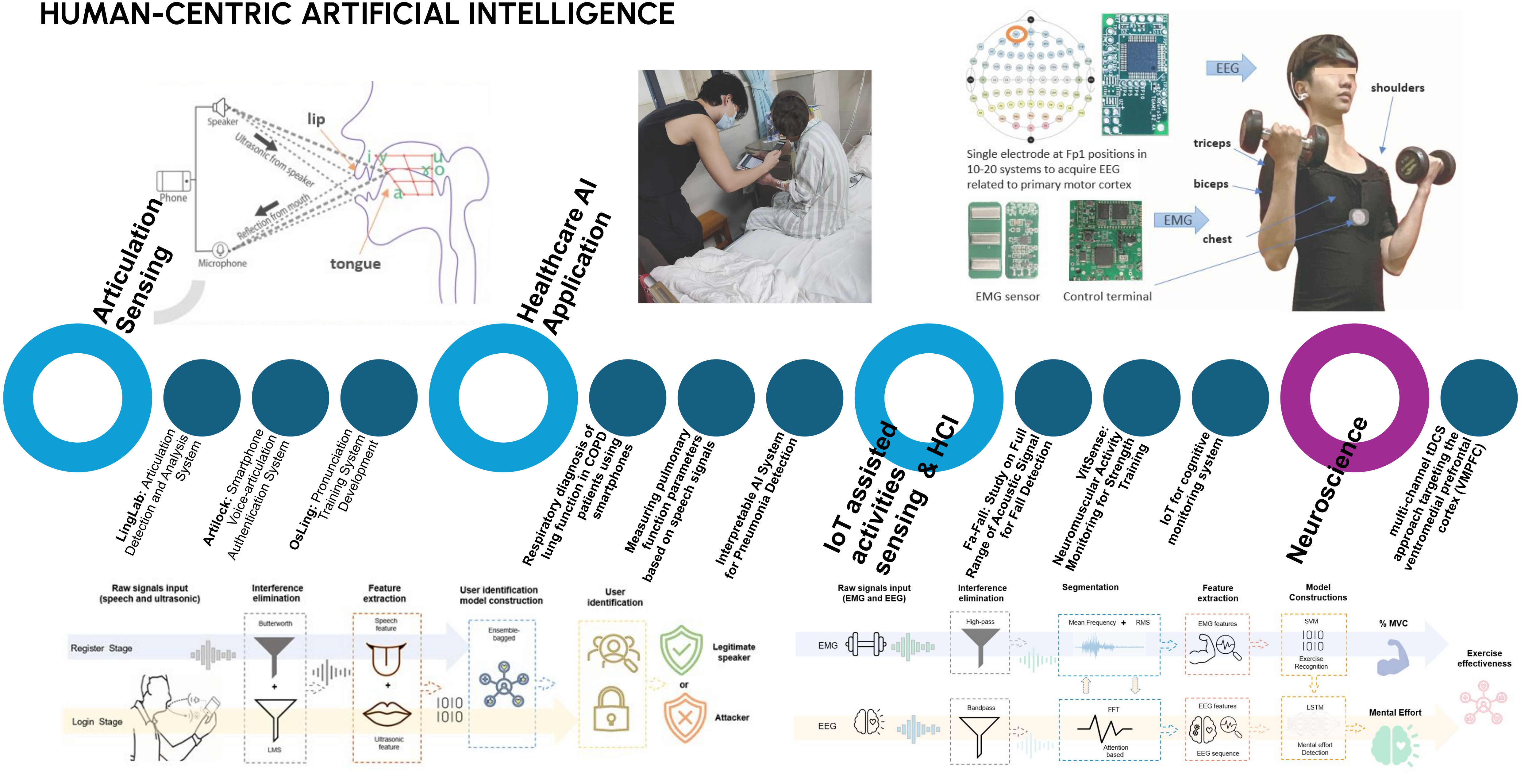
1. Articulation Verification



3. User Authentication



HUMAN-CENTRIC ARTIFICIAL INTELLIGENCE



Look for product ideas in top HCI and ubiquitous computing venues.

Use papers as inspiration for user problems, sensing patterns, interaction design, and evaluation methods.

ACM UbiComp
everyday sensing

IMWUT
wearable + mobile data

CHI
human needs + interaction

MobileHCI
mobile/wearable UX

UIST
new interaction techniques

CSCW
teams, care, work routines

themes become product signals

human behavior + sensor data + feedback loop

1. **Read for pain point** →
2. **Extract measurable signal** →
3. **Design feedback moment** →
4. **Build MVP proof**

Turn research inspiration into a startup-style hackathon idea.

Do not copy a paper. Translate a proven human problem or sensing pattern into a useful wellness product loop.

Paper theme

office posture, interruption, stress cue, daily rhythm, mobility, teamwork, attention

Product user

student, desk worker, caregiver, athlete, maker, nurse station, elderly family

Signal strategy

IMU + sound + light + distance + context + self-report event

MVP loop

sense → classify state → dashboard → feedback cue → user action

HCI theme: attention and sedentary work → Product: Focus Coach that times micro-breaks using posture, inactivity, and light.

Claim boundary: wellness awareness and behavior support only; not diagnosis, treatment, or medical decision-making.

: ตารางดำเนินงาน Hackathon

ใช้ timeline นี้เป็น operating plan กลางสำหรับทุกบ้านและทุกทีม

Sun, May 24, 2026

20.00 แต่ละบ้านส่งรายชื่อกลุ่ม เช่น EXP-1, EXP-2, ...
กลุ่มละ 8-9 คน

Mon, May 25, 2026

13:30
20.30 อ.อัสลาน office hour / ทีม infra พี่สัน / พี่ป่อง

Tue, May 26, 2026

10.00 แต่ละกลุ่มส่งรายชื่อหัวข้อตามโจทย์

13:00 Office hours: Med team

Thu, May 28, 2026

09.00-16.30 อ.ธนาให้คำปรึกษาด้านเทคนิค (Uno Q)ตามบ้าน

Fri, May 29, 2026

09.00 ปิด Hackathon และส่งสไลด์นำเสนอเข้าระบบ mysuperai

09.00-10.00 ติดตั้ง prototype demo ทุกทีมที่ห้อง To Be Number One หลังติดตั้งไม่อนุญาตให้เข้าไปแก้ไขใด ๆ

สามารถใช้ notebook ของตนเองแสดงผล Dashboard ได้